114050

Sustainable Energy Technologies Department
NIT Kurukshetra

Class Test 1
NSF (SETPC204)
Attempt all questions

B.Tech 3rd semester
Date: 24/09/2025
Time 50 Min

1. Explain the properties of Positive Real Functions [3]

2 Realise Foster-II forms of the LC impedance function [3]

$$Z(s) = \frac{(s^2+1)(s^2+3)}{s(s^2+2)}$$

3 Determine the Foster-I form of realisation of the RC impedance function. [4]

$$Z(s) = \frac{(s+1)(s+3)}{s(s+2)(s+4)}$$

4 Find the Cauer forms of the RL impedance function: [5]

$$Z(s) = \frac{2(s+1)(s+3)}{(s+2)(s+6)}$$